

How do humans learn?

Sociocultural & Sociopolitical Foundations

SYS 478. Fall 2026. Prof. Sarah Van Wart



Outline

1. XXXX
2. Sociocultural considerations
3. Socio-political considerations

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1. ????
2. **Sociocultural considerations**
3. Socio-political considerations

What is intelligence (again)?

A working definition, testable across systems: “a system's generalized capacity to process environmental data and change its behavior to maximize success” (Legg & Hutter, 2007).

Type	Description	Citation
Human Intelligence	High-level mental capability to reason, solve problems, plan for the future, and pass down knowledge.	Neisser et al. / APA Task Force (1996), "Intelligence: Knowns and Unknowns"
Animal Intelligence	The capacity to learn from experience, navigate social structures, and adaptively use tools.	de Waal (2016), <i>Are We Smart Enough to Know How Smart Animals Are?</i>
Plant Intelligence	Decentralized information processing allowing organisms to optimize growth and defense through environmental sensing.	Trewavas (2014), <i>Plant Behaviour and Intelligence</i>
DNA / Genetic	Molecular information storage and processing that can be harnessed to perform computation, via encoding, replication, and selection.	Adleman (1994), "Molecular Computation of Solutions to Combinatorial Problems," <i>Science</i>
Non-Living / AI	Mathematical capability of digital tools to parse massive data, map complex patterns, and execute actions as rational agents.	Russell & Norvig (2020), <i>Artificial Intelligence: A Modern Approach</i>

Neuroscience accounts tend to be functionalist

Biological and neuroscientific accounts tend to privilege functional explanations – mechanisms, capacities, and behavior (i.e., the brain as an information processing system) – even when they study experience or context.

- **Subjective experience** is studied (e.g., consciousness, pain, emotion, affect), – but usually through neural mechanisms.
- Some approaches are more **situated**, examining how development, environment, stress, and culture shape the brain.

What is intelligence (again)?

There is no one settled definition of intelligence, but most people consider it to be the ability to learn from experience and to adapt to, shape, and select environments.

- Intelligence isn't just “in the head.” It's distributed across tools, settings, and relationships.
- The question isn't whether a tool, setting, or relationship changes your intelligence. It always has.
- The question is: what capabilities does your environment augment?

The question isn't *whether* a tool or a setting or a relationship changes your intelligence. It always has. The question is *what capabilities does your environment augment.*

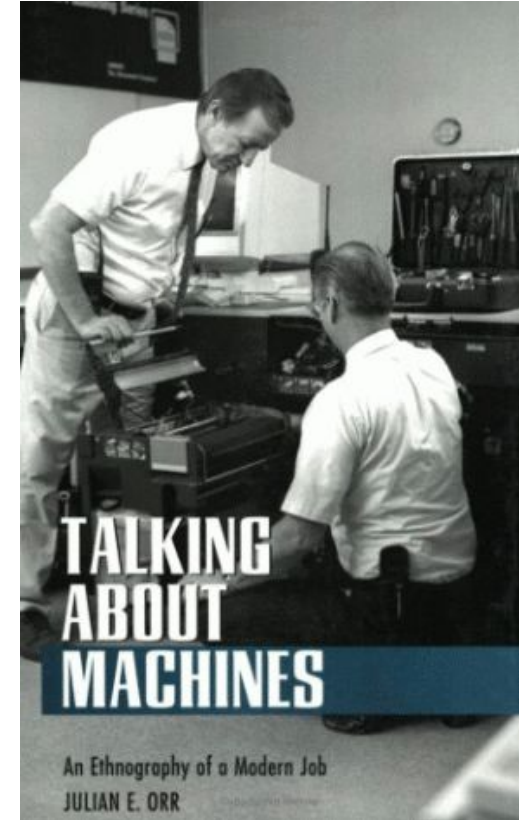
What is “Common Sense” / Tacit Knowledge?

You thoughts here...

What is “Common Sense” / Tacit Knowledge?

Examples:

- Julian Orr and the Xerox Machine Instructions
- **“Work to rule” strikes:** A strike where workers only do what is explicitly stated in their job description, performing no additional tasks beyond the minimum required, effectively slowing down operations to pressure the employer into concessions.



What is “learning”?

- The process of acquiring new knowledge, skills, behaviors, or understanding through **experiences**, **study**, or **instruction**.
- Can be conscious (like studying a new subject) or unconscious (such as picking up social cues, or through trial and error).
- How and what you learn varies drastically based on your environment – including social and cultural dimensions, the tools and resources to which you have access.
- The human mind is constantly learning. However what is learned varies greatly, depending on environmental factors.

What else matters beyond intelligence for the human experience?

Functional

What one *does*

- Intelligence
- Learning
- Memory
- Motivation
- Choice / agency
- Language
- Imagination

Experiential

How one *experiences* their existence

- Consciousness
- Emotion
- Pain / pleasure
- Desire ¹
- Sense of self
- Meaning / significance ²

Situated

The *conditions* under which one exists

- Embodiment
- Mortality
- Vulnerability
- Dependence
- Sociality
- Culture
- History
- Place / environment

1. Probably should be cross-listed with “functional”

2. Probably should be cross-listed with “situated”

Sociocultural Theory

Big idea: learning is mediated through the lens of culture and history. It always depends on the contexts in which the learner is situated. It looks different across cultures.

- What is learned and why it is learned
- How it is learned (observation, direct instruction, participation in a learning community, trial and error)
 - Tacit knowledge is informal, undocumented, and incredibly important
- What tools and resources are available in the environment

Example: Kris Gutiérrez

- People are ***made smart*** through the tools and resources they have access to in their environment.
- Being smart is relative and situated.

The question is not “How can we fix” Judith?” The question is: How do we re-configure the learning environment – with resources, tools, and other forms of assistance – so that Judith can be the smart person that she already is?”



Kris Gutiérrez, UC Berkeley

Lev Vygotsky: Influential Sociocultural Theorist

- Known for emphasizing the importance of the social and cultural dimensions of learning.
- **Zone of Proximal Development (ZPD)** – children learn best when they work on tasks slightly beyond their current abilities with guidance from more knowledgeable others.
- Importance of **language & culture** in thought development, proposing that “internal speech” helps children organize thoughts and solve problems.



Sociocultural Theory > Apprenticeship

- Definition: A model of learning involving guided participation in a community of practice (Jean Lave).
- Key Aspects:
 - Learning through observation and hands-on experience.
 - Mentorship and expert guidance.
- Examples: [apprenticeships](#) and internships.

Outline

1. What is intelligence? What is learning?
2. Behaviorism
3. Constructivism
4. Sociocultural considerations
- 5. Socio-political considerations**

Socio-Political Factors

- How societal structures and policies impact learning
- Power dynamics affecting what knowledge is valued and taught

Race, Gender Categories, and Stereotypes

- Stereotypes can influence student performance (e.g., [stereotype threat](#)) – powered by media and social media algorithms
- Different identities can get “steered” to different learning experiences – which can impact ideas around self-efficacy (what students feel like they’re capable of).

Broader Contexts

- How funding decisions affect educational resources and opportunities – like with technology, certain categories of students are chronically underserved.
- Curriculum wars:
 - Critical Race Theory & “Foundations of American History”
 - “Don’t say gay” laws
 - “New Math”
 - “Back to Basics” v. “Exploratory / Experiential Learning”
- Cultural communities of teachers often different from the cultural communities of students
 - Can lead to deficit views of certain student communities and their families.

Summary of Learning Theory

Behaviorism	Focuses on observable behaviors and the role of environmental stimuli, asserting that learning occurs through reinforcement and punishment.
Constructivism	Learners actively construct knowledge through experiences and interactions, emphasizing the importance of prior knowledge and social contexts.
Sociocultural considerations	Emphasizes the importance of social interactions and cultural contexts on learning. From this vantage point, knowledge is co-constructed through communication and collaboration.
Socio-political considerations	How broader societal structures, including power dynamics and economic factors, impact learning (e.g., beliefs about what is possible, access to material, ideational, and relational resources).

Career Discussion

Where you are with your decision about what to do after graduation?

- **Fully undecided:** you have not decided what you would like to do with your degree after graduation
- **Decided, but not taken action:** you know what you would like to do with their degree after graduation, but have not taken steps to pursue it
- **Decided and taken action:** you know what you would like to do with your degree after graduation, and have taken at least one step toward pursuing it